UNIVERSITY OF WESTMINSTER

INFORMATICS INSTITUTE OF TECHNOLOGY

Initial Draft Project Analysis

KJ Marketing

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Group A8

Team Members

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1. Introduction & Problem Understanding

1.1 Introduction to KJ Marketing

KJ Marketing, a subsidiary of the Octave Data and Advanced Analytics Division of the John Keells Group, operates a leading supermarket chain in Sri Lanka. With 22 outlets across urban and suburban regions, the company offers a diverse product range, including dry items, fresh items, and luxury items, catering to a broad customer base.

In recent years, KJ Marketing has noticed that traditional marketing strategies have become less effective in engaging their evolving customers. They aim to adopt a personalized marketing strategy tailored to individual customer preferences to address this.

1.2 Business Problem

Traditional marketing strategies have become ineffective in engaging KJ Marketing's evolving customer base. The company aims to implement a personalized marketing strategy by segmenting customers based on purchasing behaviour. Our task is to develop a machine learning model that can classify new customers into predefined segments using historical sales data. The key challenge is to classify new customers into six predefined customer segments based on their purchase behaviour. This classification will help KJ Marketing improve:

- Personalized promotions
- Targeted marketing campaigns
- Inventory management based on customer preferences

1.3 Project Objectives

This project aims to develop an analytical solution that can classify new customers into appropriate segments. Specifically, we aim to:

- Analyse historical sales data to understand purchasing patterns.
- Develop a machine learning model to classify customers into segments based on their buying behaviour.
- Evaluate and refine the model for accuracy and effectiveness.
- Provide business insights to KJ Marketing on how they can improve their marketing strategies.

By successfully classifying customers into meaningful segments, KJ Marketing can create personalized offers and optimize its marketing efforts to enhance customer engagement and sales.

2. Preliminary Data Analysis (EDA)

2.1 Dataset Overview

The datasets provided for this project include:

- **train.csv** – Used for model training and testing.
  - customer_id: Unique customer identifier
  - outlet_city: Outlet location
  - luxury_sales: Average monthly spending on luxury items
  - fresh_sales: Average monthly spending on fresh items
  - dry_sales: Average monthly spending on dry items
  - cluster_category: Predefined customer segment (target variable)
- **test.csv** – Used to evaluate the trained model.
  - Same features as train.csv excluding the cluster_category

2.2 Initial Observations

- **Missing Values:** Need to check for null values and decide on imputation strategies. Missing values are identified in all columns but mostly in sales columns.
- **Duplicates:** Checked to Identify and remove any redundant records and no duplicates were found.
- **Outliers:** Detected extreme sales values that could distort the model.
- **Format Issues :** There are problems with formatting and data types in some columns that need to be dealt with.

2.3 Planned Data Pre-processing Steps

- Handle missing values using mean/median imputation or dropping if insignificant.
- Formatting columns into proper data types. Ensuring Numeric columns are formatted into Number with a specific decimal place.
- Removing Outliers using Interquartile range to ensure data integrity.
- Normalizing and Standardizing numerical features to improve model performance using pre-processing techniques from sklearn.preprocessing.
- Encode categorical variables appropriately using ‘LabelEncoder().
- Splitting data in Train dataset into 70% training and 30% validation sets for model evaluation.

3. Project Plan & Team Roles

3.1 Work Breakdown Structure (WBS)

Task	Description	Timeline	Member
Task Allocation	Assigning tasks to team members	15 Feb – 17 Feb	Project Manager
Initial Analysis and Project Plan	Creating project plan and scope	18 Feb – 24 Feb	Project Manager
Data Exploration	Exploring dataset for insights	25 Feb – 27 Feb	Data Analyst
Data Pre-processing	Cleaning and preparing data	28 Feb – 02 Mar	Data Engineer
Feature Engineering	Creating new features from data	03 Mar – 07 Mar	Data Scientist
Customer Segmentation	Clustering customers based on data	08 Mar – 11 Mar	ML Engineer
Model Tuning and Validation	Improving model performance	11 Mar – 14 Mar	ML Engineer
Model Interpretation	Understanding model predictions	15 Mar – 17 Mar	Business Analyst
Draft Report for Client	Preparing first draft of report	18 Mar – 20 Mar	Report Writer
Presentation Preparation	Creating presentation slides	21 Mar – 24 Mar	Presentation Lead
Final Report	Finalizing project report	25Mar – 28 Mar	Report Writer
GitHub Repository	Organizing code repository	29 Mar – 31 Mar	GitHub Manager
Submit Final Deliverables	Submitting all project materials	3 April	Project Manager

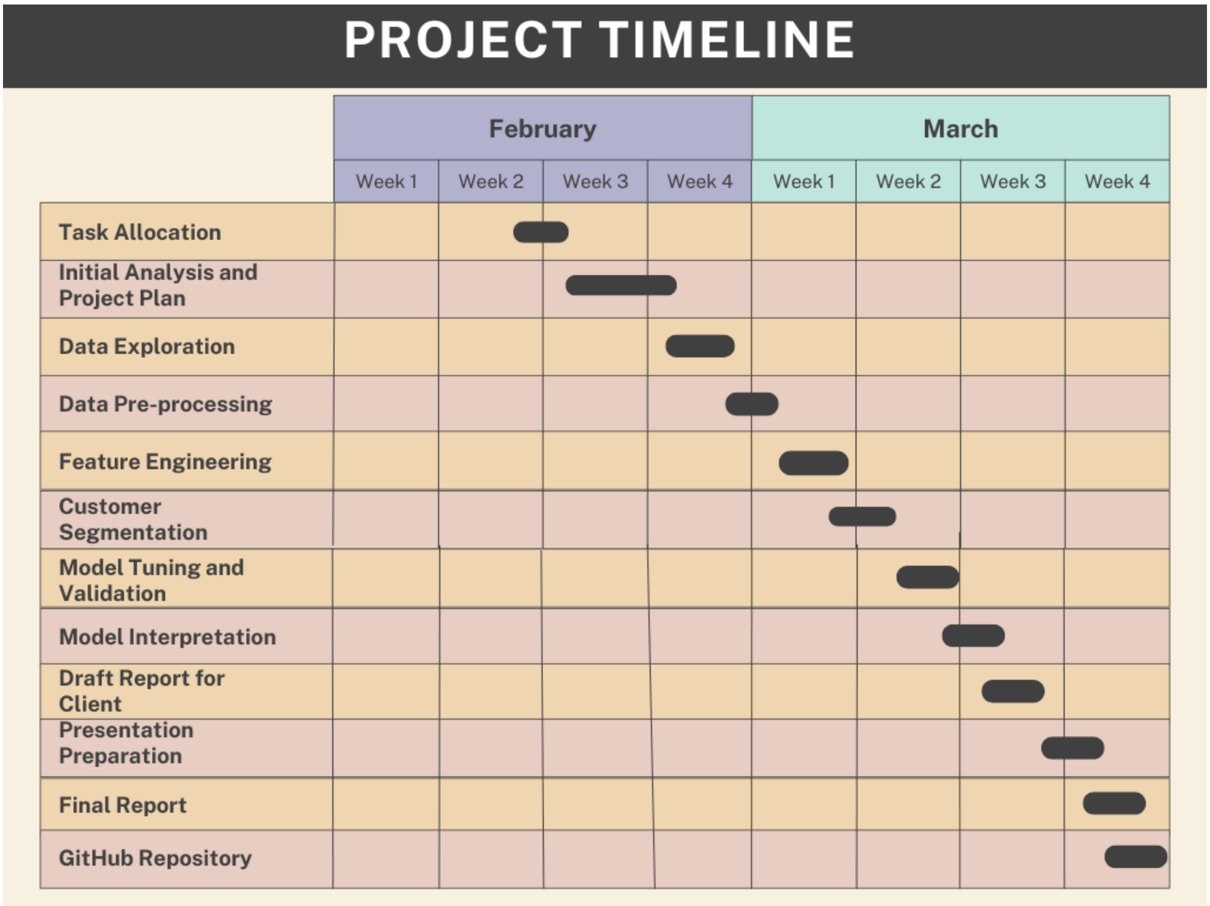
Nethmi Maleesha – Team leader/ Project Manager / Data Scientist

Sithuli Kothalawala – Data Engineer / Report Writer

Shahma Silmy – Data Analyst / Business Analyst / Presentation Lead

Ruchintha Dias – ML Engineer / GitHub Manager

3.2 Gantt chart



4. Potential Risks & Mitigation Strategies

Risk	Mitigation Strategy
Data quality issues (missing values, outliers)	Implement robust data cleaning methods
Model overfitting	Use regularization techniques and cross-validation
Lack of interpretability in ML models	Choose explainable AI methods such as decision trees
Team coordination challenges	Weekly progress meetings and task assignments